Experimental supplement: COMNET-D-26-03761

The six formal batches contain 340 completed two-flow runs: 60 each for the Ubuntu in-tree baseline, separate BBRv1 module, BBR2 and BBRv3, 80 E3 runs, and 20 artificial-loss runs. Each condition has five repetitions. The original configurations and raw logs are supplied without alteration.

Primary metrics use receiver end.sum_received.bits_per_second in the iperf client JSON, divided by 1e6. Total goodput is x1+x2; Jain=(x1+x2)^2/[2(x1^2+x2^2)]; second-flow share=x2/(x1+x2). Average these quantities across runs. In E2 flow 1 is BBR, flow 2 CUBIC; mean BBR share=1-mean second-flow share. experiment_condition_variability.csv supplies means and sample standard deviations for all 68 conditions. verify_experiments.py independently reproduces rates and metrics from the raw JSON; run it from the extracted package root.

The original batch scripts retain their analysis paths and are historical reproduction materials. The verification script is the portable numerical check; figure scripts are in the manuscript source package. Controller source archives are identified by manifests where available, but are not redistributed here. Git commit metadata was not available in those archives. Kernel environment and the BBRv1 compatibility patch constrain causal version comparisons.

Important interpretation: configuration warmup_s=30 and analysis_start_s=30 do not describe the primary reported result window, which includes the whole approximately 180-second receiver transfer. measurement_window_audit_20261012 contains the separate whole-interval 30–180-second sensitivity analysis. RTT probes are early sequential ping samples, and queue snapshots precede transfer completion; zero snapshot counters cannot establish zero full-run drops. In the 50-Mbps E3 cases the actual Q/BDP targets are 4 and 16, despite frozen labels 1 and 4. At 200 Mbps the actual ratios are 1 and 4. Raw labels remain unchanged. Artificial loss is configured on a separate receiver access link at 0 or 0.1%, without a retained random seed; it does not model correlated wireless errors.

No new experiments were run during final editing. These materials support descriptive comparisons under the documented settings, not universal applicability thresholds or an isolated causal ranking of controller versions.

The targeted revision adds an analytical assessment and a retrospective E1-to-E3 transfer evaluation. Fifteen equal-RTT BBRv3 E1 runs calibrate log-queue interpolation; E3 outcomes are not fitted. Four input-matched 200-Mbps cases and four corrected 50-Mbps cases are evaluated separately. Normalized-goodput/Jain errors are compared with a constant calibration baseline, retaining the worse Jain result for the corrected group. The analysis was designed after archived outcomes were available and is not preregistered. All run-level decisions, condition variability, a utilization/fairness threshold grid and portable scripts are in targeted_revision. The chosen service requirements are illustrative, not universal physical thresholds.